
Software Requirements Specification

for

Sakthi Auto - Industry innovation Web Portal

Version 1.0 approved

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1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for the **Sakthi Auto – Industry Innovation Web Portal**. This web-based platform is designed to bridge the gap between industry and academia by facilitating real-time innovation challenges, idea submissions, and consultancy tracking.

1.2 Scope

The software will function as a collaborative platform where:

- **Sakthi Auto** can upload industrial problem statements and evaluate submissions.
- **Academic Institutions** can register, view challenges, and submit Proof of Concept (PoC) or theoretical solutions.
- The system includes dashboards for tracking the status of ideas from submission to consultancy engagement.

1.3 Definitions and Acronyms

- **SACL**: Sakthi Auto Component Limited.
- **SPOC**: Single Point of Contact (Institution Admin).
- **PoC**: Proof of Concept.
- **SMTP**: Simple Mail Transfer Protocol (used for notifications).
- **SRS**: Software Requirements Specification.

2. Overall Description

2.1 Product Perspective

The product is a standalone web application designed to replace manual communication methods between industry and academia. It addresses the gap where students often work on outdated topics by providing a direct channel for real-world industrial problems.

2.2 User Classes and Characteristics

The system shall support the following user roles with specific access privileges:

- **Admin (SACL HR):** Has full control over the system, including problem uploads, user management, and dashboard monitoring.
- **Evaluator (SACL Department Heads):** Responsible for Problem uploads from thier domain or department, reviewing submissions, scoring ideas, and adding technical comments.
- **College SPOC (Institution Admin):** Acts as the central coordinator for the college, receiving alerts and managing team submissions.
- **College Team (Faculty/Students):** End-users responsible for generating ideas and uploading solutions.

2.3 Operating Environment

- **Client Side:** The application shall be accessible via standard web browsers (Chrome, Firefox, Edge) on desktop and mobile devices.
- **Server Side:** Hosted on AWS (Amazon Web Services).
- **Database:** MS-SQL for relational data storage.

3. System Features (Functional Requirements)

3.1 Module 1: College Onboarding and Authentication

- **REQ-1.1:** The system shall allow Engineering Colleges to register and create a profile.
- **REQ-1.2:** During registration, the system must capture department details, domain interests, and key contact information.
- **REQ-1.3:** The system shall provide secure login credentials for the College SPOC upon successful registration.
- **REQ-1.4:** For each college, a single SPOC ID will be created.
- **REQ-2.3:** The system shall display active challenges on the College Dashboard.

3.2 Module 2: Problem Statement Management

- **REQ-2.1:** The **Admin** shall be able to create, edit, and delete industrial problem statements and **assign** specific problem statements to an Evaluator (Department Head).

- **REQ-2.2: Evaluators** shall be able to create, edit, and upload new problem statements relevant to their specific domain or department.
- **REQ-2.3:** Upon posting a new problem (by either Admin or Evaluator), the system shall trigger the Automated Notification System to send emails to all registered College SPOCs.
- **REQ-2.4:** The system shall display all active challenges on the College Dashboard, regardless of whether they were created by an Admin or an Evaluator.

3.3 Module 3: Idea Submission

- **REQ-3.1:** Authenticated College Teams shall be able to upload solutions in the form of Proof of Concept (PoC) or theoretical documents.
- **REQ-3.2:** The system must generate an automatic acknowledgment receipt (Email) for the college upon successful submission.
- **REQ-3.3:** Users shall be able to track the current status of their submission (e.g., Submitted, Under Review) via the dashboard.

3.4 Module 4: Evaluation and Scoring

- **REQ-4.1:** Evaluators (SACL Dept Heads) shall have access to view submissions assigned to their specific domain.
- **REQ-4.2:** The system shall allow Evaluators to input scores and update the status of an idea to "Shortlisted," "Under Review," or "Rejected".
- **REQ-4.3:** The system shall update the College Dashboard in real-time to reflect the evaluation results.

3.5 Module 5: Consultancy Engagement Tracking

- **REQ-5.1:** The system shall provide a specific dashboard view for ideas marked as "Shortlisted".
- **REQ-5.2:** The Admin shall be able to log details regarding prototype development and consultancy collaboration outcomes.
- **REQ-5.3:** The system must record the entire engagement history for analytics purposes.

4. External Interface Requirements

4.1 User Interfaces

- The Front-end shall be developed using **React.js with Tailwind CSS** to ensure a responsive and user-friendly design.
- Dashboards must be provided for both Sakthi Auto (Admin) and Colleges.

4.2 Software Interfaces

- **Runtime Environment:** Node.js.
- **Server Logic:** Express.js.
- **Database:** MS-SQL.
- **Email Service:** SMTP (Simple Mail Transfer Protocol) for notifications.

4.3 Communications Interfaces

- The system shall utilize HTTPS for secure communication over the internet.
- The system shall interface with **AWS (Amazon Web Services)** for hosting and deployment.

5. Non-Functional Requirements

5.1 Performance

- The system shall support real-time updates for problem posting and status changes.
- The database shall be optimized to handle records of participation and performance analytics.

5.2 Security

- Access to the Admin Dashboard shall be strictly restricted to authorized Sakthi Auto personnel.
- The system must ensure data integrity for all uploaded PoC documents and consultancy records.

5.3 Maintainability

- The source code shall be version-controlled using **Git & GitHub** to facilitate collaboration and future updates.

5.4 Design Constraints

- The UI/UX design and wireframing must be conducted using **Figma** prior to development.